Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/03/2024
PI Name:	Zubair, Abba M.D., Ph.D.		
Project Title:	Role of Mesenchymal Stem Cells in Microgravity Induced Bone Loss		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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City:	32224-1865	**State:**	FL
Zip Code:	32224-1865	**Congressional District:**	4
Comments:			
Project Type:		**Solicitation / Funding Source:**	03-OBPR-02
Start Date:	02/01/2021	**End Date:**	01/31/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 01/31/2026 per NSSC information (Ed., 12/17/24) NOTE: End date changed to 01/31/2025 per F. Hernandez/ARC (Ed., 6/18/24) NOTE: Period of performance updated to 2/1/21-1/31/24 (from 12/1/20-11/30/23) per F. Hernandez/ARC and NSSC information (Ed., 2/2/21)		
Key Personnel Changes/Previous PI:	DECEMBER 2022 Dr. Mannie Saunders is no longer collaborating in this research project because she has retired from research and closed her lab at University of Akron.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC21K0299		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Bone loss resulting in low bone mineral density as observed in osteopenia and osteoporosis increases the risk of fracture. Bone loss is a major problem at Earth gravity (1 g) and is aggravated by microgravity in space, especially during long duration space flight. Exercise slows bone loss in microgravity; but even when astronauts adopt a rigorous exercise regimen, they continue to lose about 1 – 2% of their bone mass a month. While there is a vast body of knowledge on bone homeostasis under normal gravity conditions, the mechanisms of bone loss in microgravity are poorly understood and remain to be explored. It is not yet fully determined what signals allow bone tissue to adapt to a weightless environment compared to an Earth environment. Conceptually, altered biomechanical stimuli in microgravity may directly affect osteoblasts, osteoclasts, and/or their precursors, while other physiological factors such as hormone levels or poor nutrition could contribute to bone loss in microgravity. The central hypothesis of our proposal is that microgravity alters the activity of mesenchymal stem cells, which are the precursors of bone forming osteoblasts, and their secretion of cytokines and growth factors that promote osteogenesis and osteolytic activity. Specifically, we postulate that microgravity may decrease the mesenchymal stem cells (MSCs) number and disturb the secretion of trophic factors that impede bone homeostasis by reducing new bone formation and/or increasing bone resorption in microgravity. Bone homeostasis requires a complex interplay between mesenchymal osteoblasts and hematopoietic osteoclasts and is controlled by multiple cytokines, including the pro-osteolytic factors interleukin (IL)-1 and tumor necrosis factor a (TNFa), and the anti-osteolytic factors transforming growth factor ß (TGFß), IL-4, and IL-10 that are pro-osteogenesis. Some cytokines such as IL-6 pleiotropically affect bone homeostasis. Our preliminary results show that microgravity culturing of MSCs at the International Space Station (ISS) does not significantly alter MSC proliferation, but alters their cytokine and growth factor secretion with an associated increase in the secretion of pro-osteolytic cytokines. We and others have shown that IL-6 secretion by MSC is very critical in MSC functioning. We propose that microgravity skews MSC secretomes towards osteolysis and that IL-6 plays a central role in the perturbation of the homeostatic balance. IL-6 activity is very critical in bone homeostasis, but IL-6 effects depend on how its signal is transmitted within its target cell. IL-6 acts via the classic signaling pathway by binding to its membrane bound receptor (IL-6R) and homodimeric glycoprotein 130 (gp130) proteins. In addition, IL-6 can act via its soluble receptor (sIL-6) coupled with gp130 protein to act on non-IL-6R expressing target cells as part of a non-canonical trans-signaling pathway. IL-6 induces osteogenesis and anti-osteolytic activities through its classical signaling pathway and induces osteolysis via its trans-signaling pathway. We postulate that microgravity provokes MSC induced bone loss and suppresses new bone formation via activation of the IL-6 trans-signaling pathway and blockade of the classic pathway. The rationale for our proposed studies is that a biological understanding of the IL-6 signaling pathways in MSC induced bone loss may permit pharmacological activation or blockage of the classical and trans-signaling pathways as a new and innovative approach to management of bone loss both in space and on Earth. Our long-term goal is to understand the mechanism of MSC induced bone loss under microgravity conditions and to develop safe and effective therapy for bone loss. The principal experimental objectives are to determine how IL-6 signaling pathways contribute to MSC induced bone loss under microgravity conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The rationale for our proposed studies is that a biological understanding of IL-6 signaling pathways and its downstream effects in MSC induced bone loss, may permit pharmacological activation or blockage of the classic and trans-signaling pathways as a new and innovative approach to management of bone loss both in space and on Earth. Our long-term goal is to understand the mechanism of MSC induced bone loss under microgravity conditions and to develop a safe and effective therapy for bone loss.
Task Progress:	Our project is studying how the lack of gravity in space changes the way mesenchymal stem cells (MSCs) behave and look, and how these changes can lead to bone loss. We also want to understand the role of a specific signaling pathway, IL-6, in this process. We have two space experiments planned: MABL-A and MABL-B. The MABL-A experiment was sent to the International Space Station (ISS) on a supply mission on January 30, 2024. Astronauts on the ISS took care of the samples and we did the same experiment here on Earth for comparison. The samples returned to Earth on April 30, 2024. We have studied these samples and started analyzing their genetic material. At the same time, we conducted a similar experiment using a device that mimics microgravity, and we are currently analyzing those samples too. Our second experiment, MABL-B, focuses on how microgravity affects bone health and the role of IL-6 in bone loss. We have finished all the necessary tests on Earth. The final verification test was completed in early November 2024, and we are now preparing for the next phase. This experiment is scheduled to go to space in June 2025.
Bibliography Type:	Description: (Last Updated: 07/14/2026)
Articles in Peer-reviewed Journals	Ghani F, Zubair AC. "Discoveries from human stem cell research in space that are relevant to advancing cellular therapies on Earth." npj Microgravity. 2024 Aug 21;10:88. Review. https://doi.org/10.1038/s41526-024-00425-0 ; PMID: 39168992; PMCID: PMC11339457 , Aug-2024